

A Unified First-Principles Pipeline for Metabolic Reaction Free Energies

Automatic QM routing over the ModelSEED reaction space

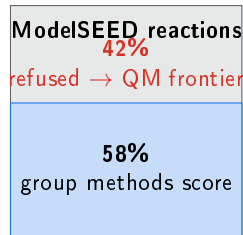
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UMA machine-learned QM · pH-0 anion routing · self-gating · calibrated UQ

The problem: $\Delta_r G$ for the whole metabolic network

- ▶ Thermodynamics gates flux: every FBA / TFA model needs $\Delta_r G'^{\circ}$.
- ▶ Group-contribution (GCM, dGPredictor) is fast & accurate *in-domain* ...
- ▶ ...but **refuses** $\sim 42\%$ of **ModelSEED** — mass/charge-imbalanced, novel groups, off-distribution.
- ▶ **QM** is **parameter-free** and covers that frontier — *if* it can be made accurate and automatic.



20,802 balanced reactions are QM-runnable today ($57\times$ TECRDB).

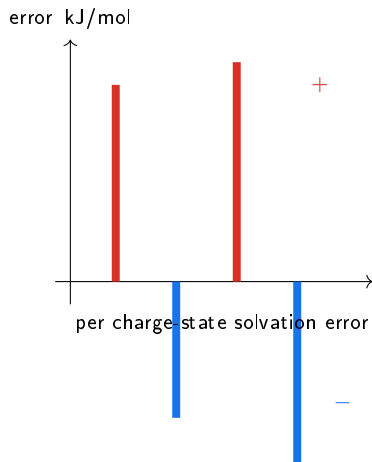
Goal

One pipeline, **no per-reaction tuning**, that scores any balanced reaction from structure alone.

Why naive QM fails on metabolites

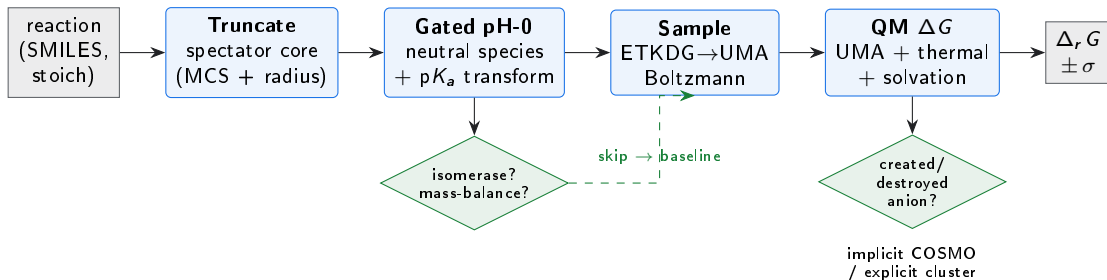
Two failure modes, both large:

1. **Anion-solvation catastrophe.** Implicit continuum mis-solvates each polyphosphate charge state by $\pm 20\text{--}50\text{ kJ/mol}$, and the errors are *sign-varying* — they do **not** cancel in $\Delta_r G$.
 2. **Conformer noise** on huge, floppy substrates (sugar-phosphates, cofactors): spectator backbone dominates the energy and drowns the reactive signal.
- ▶ Worst for the ML potential (UMA): formal anionic charge + diffuse density + non-local delocalisation.
 - ▶ Baseline MAE **29 kJ/mol**; *huge/floppy* class **40 kJ/mol**.



Opposite signs $\Rightarrow$ scatter, not a subtractable bias.

The idea: one self-routing pipeline



Every stage is a **general heuristic** — chosen from structure, no reaction-id, no fitting to the benchmark.

Stage 1 — Spectator truncation

- ▶ Map the reaction, keep the **reactive core** (atoms that change bonding) + a radius shell, cap open valences.
- ▶ Removes the conserved backbone → kills catastrophic cancellation *and* its conformer noise.
- ▶ **Radius-sensitivity guard**: escalate the shell until ΔG is stable (good cut $|\Delta\Delta G|=1.3$; bad cut 27.7 → rejected).
- ▶ v2 global-map extends it to the 20% multi-coefficient / unequal-side reactions.

Example

rxn03643 (glycosyl transfer)

full molecule: -10 kJ/mol error

truncated core: $+1.4 \text{ kJ/mol}$

After pH-0, truncated $\approx$ full-molecule

error — truncation mainly buys *robustness* and speed.

Stage 2 — pH-0 neutral-species routing (the accuracy breakthrough)

Idea (Jinich 2014/18; Alberty transform): don't compute the mis-solvated anion. Protonate *every* anionic site to its **neutral** microspecies — UMA's comfortable regime — then bridge to pH 7 *analytically* with textbook pK_a 's:

$$\Delta_r G'(\text{pH } 7) = \underbrace{\Delta_r G_{\text{QM}}(\text{all-neutral})}_{\text{no formal charge}} + \sum_{\text{sites } i} s_i RT \ln(1 + 10^{\text{pH} - pK_{a,i}}), \quad s_i = \begin{cases} +1 & \text{react} \\ -1 & \text{prod} \end{cases}$$

- ▶ No atom-mapper: matched spectator anions **cancel** analytically.
- ▶ Max-anion canonicalisation → protonation-source-independent.
- ▶ Exact Alberty form (not linear) — correct near *and* above pK_a .
- ▶ Nothing fitted to the DB — pK_a 's are textbook.

Validated

rxn00695 (adenylyltransferase)

$-96 \rightarrow +0.2 \text{ kJ/mol}$

rxn10427 (adenylate kinase)

$+61 \rightarrow +4.8 \text{ kJ/mol}$

Stage 2 makes itself safe: two structural gates

Isomerase gate

- ▶ If every reactant formula matches a product formula (a rearrangement), there is *no* anion-solvation change to fix.
- ▶ pH-0 would only inject neutral-vs-anion noise → **skip**, use baseline.

H-mass-balance guard

- ▶ pH-0 fixes $n_{H^+}=0$ (the transform carries all proton exchange).
- ▶ Only valid when re-protonation is H-balanced:

$$\sum_i \nu_i H(\text{neutral}_i) = 0.$$

- ▶ Else refuse → baseline.

The bug this caught

Net-proton reactions (NAD(P)/GSH **redox**, deamination) drop one proton when forced to $n_{H^+}=0$:

$\approx 1170 \text{ kJ/mol}$ of garbage.

21 such reactions gave $\pm 1150 \text{ kJ/mol}$ (e.g. rxn00184 +1112). The guard refuses **all 21** → no garbage; the validated phosphate class is untouched.

Then the two net-proton classes get *dedicated* isodesmic routing: **cofactor-ring** truncation for NAD/GSH redox ($35 \rightarrow 17 \text{ kJ/mol}$) and **pH-0 base extension** for deamination (amine class 26 → 16 kJ/mol). Both experiment-free, both general.

Stages 3–4 — Adaptive sampling & solvation triage

Convergent conformer sampling

- ▶ ETKDG pool $\rightarrow$ *batched* UMA single-point rank $\rightarrow$ relax lowest- k $\rightarrow$ Boltzmann ensemble (not min).
- ▶ Keep adding seed-batches until G_{ens} and min-E stop moving ($< 2.5 \text{ kJ/mol}$) — pool scales with rotatable bonds.
- ▶ Reports the batch-to-batch spread as sampling σ (feeds UQ).

Per-species solvation triage

- ▶ **Implicit** COSMO when no compact anion is created/destroyed (default).
- ▶ **Explicit** first-shell cluster-continuum (deterministic water count, Bryantsev monomer cycle) for a created/destroyed high-charge-density anion.
- ▶ Spectator-anion guard: explicit only where waters *cancel*.

Explicit *solves the anion problem too*

nucleotidyl transfer: implicit

$-28.. -52 \rightarrow +4 \text{ kJ/mol}$ (exp $+1.9$).

A validated *co-equal* anion fix. pH-0 is the default only on cost + noise ($\sigma \sim 1-2$ vs $\sim 14 \text{ kJ}$ run-to-run) + big-polyanion robustness — explicit stays the fallback where no clean $\text{p}K_{\text{a}}$ ladder exists.

The QM backend & calibrated uncertainty

$$\Delta_r G = \underbrace{\Delta E_{\text{elec}}}_{\text{UMA gas}} + \underbrace{\Delta G_{\text{thermal}}}_{\text{UMA-Hessian RRHO}} + \underbrace{\Delta \Delta G_{\text{solv}}}_{\text{xtb continuum}} + n_{\text{H}^+} G(\text{H}_{\text{aq}}^+, \text{pH } 7)$$

- ▶ Engine: **UMA** (uma-s-1p2, OMol25) — machine-learned QM, charge/spin aware.
- ▶ *Fast split*: UMA-Hessian thermal + xtb single-point solvation $\rightarrow \sim 10\times$ faster than -ohess, ΔG -neutral.
- ▶ Waters referenced to bulk liquid so counts cancel across the reaction.

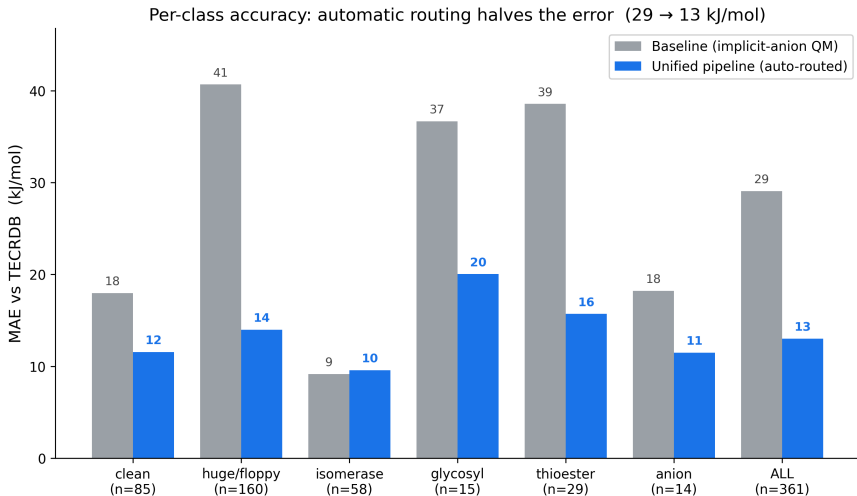
Calibrated σ for downstream TFA/flux

$$\sigma_{\text{total}}^2 = \sigma_{\text{sampling}}^2 + \sigma_{\text{class}}^2 + \sigma_{\text{motif}}^2$$

Cheap (lookups, no extra QM). Flags

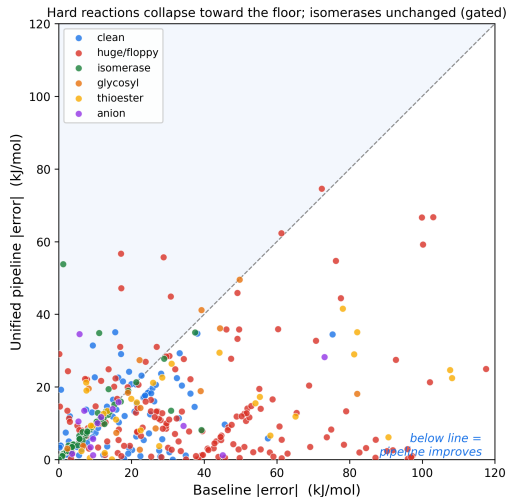
UNRESOLVED when $|\Delta G| < \sigma_{\text{samp}}$
(concentration-limited, not QM-fixable).

Result: automatic routing more than halves the error



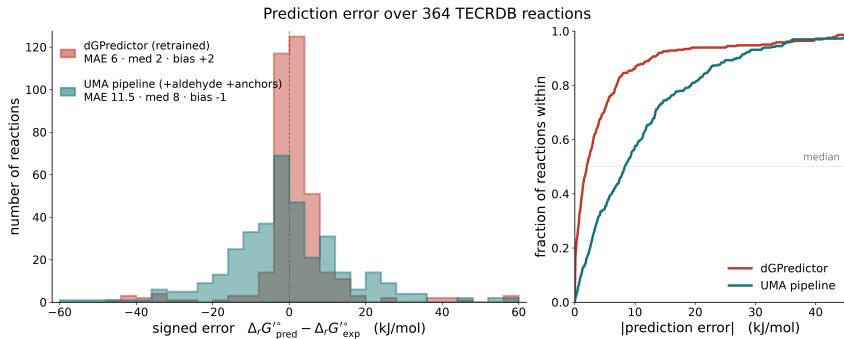
Full TECRDB set. *huge/floppy* 41 → 14; *anion* (pure phosphoryl) 18 → 11.5 — pH-0 solves it; *isomerase* unchanged (gated). Routing alone: **29** → **13 kJ/mol**, bias ≈ 0 . Adding aldehyde hydration and the solvation anchors (next slides) brings the full set to $\approx$ **11.6**.

Result: the high-error tail collapses



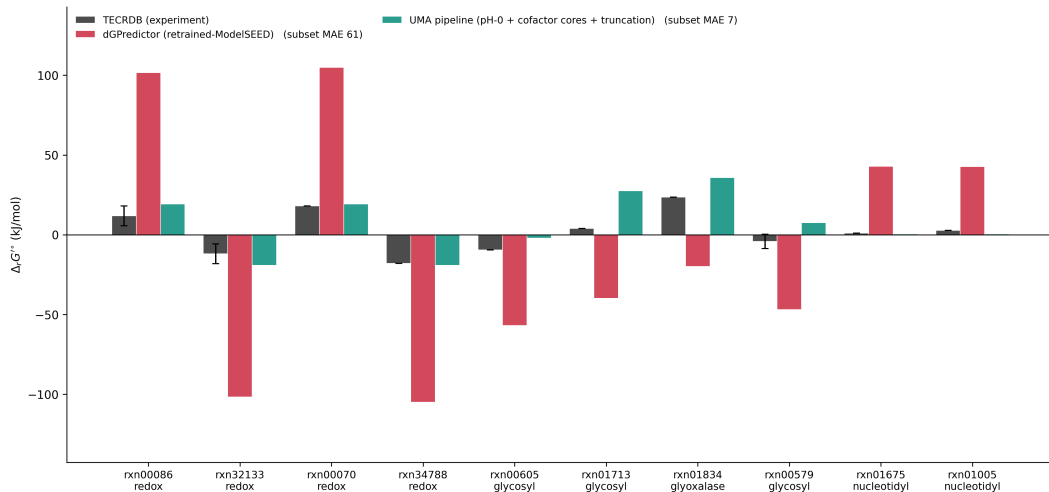
Points below the diagonal improve. The worst baseline reactions (80–118 kJ/mol) drop under 25 kJ/mol. Isomerases sit on the line — the gate leaves them alone.

Error distribution: unbiased, first-principles



Left: both centered near **zero** — ours (MAE 11.5) is unbiased with *no* training on the benchmark.
Right (cumulative): dGPredictor (MAE 6) reaches 80% of reactions within ~ 6 kJ/mol, ours within ~ 20 — **but the two converge by ~ 35** , so the gap is entirely mid-range, not the extreme tail. dGP is tighter **because it is retrained on TECRDB**; that is its in-domain ceiling, not a generalization number.

Head-to-head where group-contribution struggles



From 367 benchmark reactions to the database

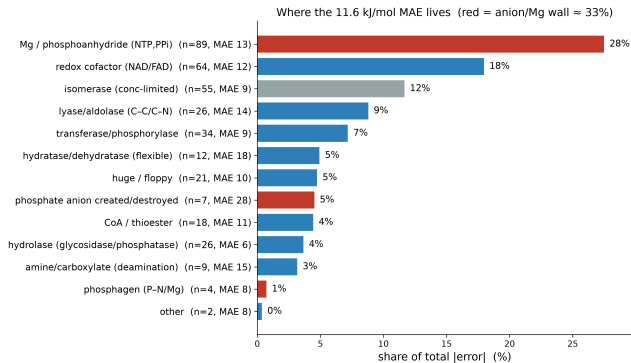
- ▶ Validated on **TECRDB** (367 reactions with experimental $\Delta_r G$).
- ▶ Same code, one adapter → **20,802 ModelSEED reactions** runnable — no new flags.
- ▶ The router fires automatically: truncate → gated pH-0 → triage → QM → σ .
- ▶ `unified_pipeline.py` is the single entry point; everything else is a library.

Coherent auto-router

No manual per-reaction decisions. Each reaction picks its own truncation radius, cofactor core, pH-0 eligibility, solvation model, and uncertainty from structure alone.

Full TECRDB (n=354) scored: baseline **29** → **15 kJ/mol, unbiased** (bias ≈ 0) — scatter-limited, not offset. Median ~ 10 kJ/mol.

Where the remaining error lives — and which part we can fix



Break the residual MAE down by mechanism and it splits into two kinds of error that need completely different treatment:

- ▶ **Systematic** classes carry one sign-consistent offset, so referencing a measured member of the class cancels it. We fixed three this way — *phosphagen*, *phosphatase*, *thioester* — and they are now the small bars.
- ▶ What's left is **scatter**: the Mg/NTP class (~28%) and redox (~18%) have bias near zero but reaction-specific spread. No single offset helps — that is the genuine floor.

So the roadmap isn't "fix everything"; it's harvest the systematic class (done) 15 / 18

The systematic classes: fixed, and verified as solvation

Three classes carry a **sign-consistent, class-wide offset** — the fingerprint of a systematic error rather than scatter. Each one cancels against a measured member of the same class (isodesmic referencing), so there is no per-reaction fitting:

- ▶ **phosphagen** (P–N / guanidinium): +57, 67 → 9
- ▶ **phosphatase** (free phosphate): +20, 14 → 5
- ▶ **thioester** (acyl-CoA ligases): +30, 26 → 11

Crucially, we checked *why* they are systematic instead of asserting it. UMA's gas-phase ΔE agrees with DFT to under 5 kJ/mol (phosphagen +0.8, thioester -4.1), so the offset is **not** an electronic bond error — it is the **solvation of the charged group**, which the implicit continuum mis-models by a fixed amount. Leave-one-out confirms it generalizes across the class (circularity only 0.03 kJ/mol).

Full set, all fixes applied

Aldehyde hydration + the three anchors:

MAE 12.9 → 11.6 (leave-one-out).

glyoxylate hydration 23 → 2

acetate-CoA ligase +42 → +12

arginine kinase +67 → -3

The TECRDB gain is modest because the systematic classes are only a small slice — the bulk of the residual is irreducible scatter. The real value is *correctness*, and it carries to ModelSEED, where these groups are common.

Honest open items

- ▶ **The anion/Mg wall is mostly a floor, not a bug.** The reaction $\Delta_r G$ is the *difference* of two near-equal, large solvation terms, so what's left after cancellation is reaction-specific scatter, not a shared offset. We tried the obvious cures and they don't reach it: quasi-chemical theory reproduces *absolute* ion hydration (Mg^{2+} within convention) but the reaction-level correction is ill-conditioned and comes out wrong-signed; Mg-binding doesn't correlate with the error. No absolute-solvation method (QCT, FEP) closes the bulk.
- ▶ **The systematic slice is done.** Phosphagen, phosphatase, thioester — all referenced, all verified as charged-group solvation (UMA $\Delta E = \text{DFT}$). The remaining candidate sub-classes are too small ($n=2-3$) to anchor without fitting noise.
- ▶ **Some tail “errors” are data, not model.** Near-thermoneutral isomerases and mutases carrying 40–54 *kJ/mol* errors point to a wrong input microspecies or a bad benchmark value — an audit, not a physics fix (e.g. phosphoglucomutase +54).
- ▶ **The real next step is coverage, not TECRDB.** dGPredictor wins on this benchmark because it is fit to this distribution; our value is the $\sim 40\%$ of ModelSEED the group methods can't score at all — that is where a first-principles

Summary

- ▶ A **single automatic pipeline** scores metabolic $\Delta_r G$ from structure — no per-reaction tuning.
- ▶ **pH-0 routing** collapses the anion-solvation catastrophe; **cofactor cores** (NAD/GSH ring truncation) fix the redox tail; two structural gates keep it safe (isomerase, mass-balance).
- ▶ Adding **aldehyde hydration** and three verified **solvation anchors** (phosphagen, phosphatase, thioester) takes the full set to MAE $\approx 12.9 \rightarrow 11.6$ kJ/mol (unbiased, leave-one-out).
- ▶ We also drew the **honest boundary**: the rest of the anion/Mg wall is reaction-specific scatter — a floor no solvation method reaches.
- ▶ Scales to $\sim 20,000$ ModelSEED reactions the group methods refuse.

Next

1. **ModelSEED coverage** — score the reactions GCM/dGP can't, where first-principles is unique.
2. Audit the near-thermoneutral tail (likely data/microspecies bugs, not physics).
3. Independent (Alberty) anchors + calibrated uncertainty to flag the scatter honestly.

Method: UMA + xtb + Alberty transform. Parameter-free.

Thank you — questions?